

Test on Introduction to object oriented programming concepts, Introduction to JAVA, Elementary concepts of objects and classes, values and data types, Operators in JAVA, Input In JAVA, Mathematical Library Methods, Conditional Constructs, Iterative Constructs (for loop, while loop(Snippets only))

Full Marks: 100
Section A

MCQ: Write the correct option

20x1 = 20

1. JAVA is an/a

- a) Object Oriented Programming Language b) Procedure Oriented Programming Language
- c) Structure Oriented Programming Language d) Low Level Language

2. Procedure Oriented Programming Language follows

- a) Top Down Approach b) Bottom Top Approach
- c) Distinct Approach d) None of them

3. Which of the following is an Unary Operator ?

- a) && b) ||
- c) ! d) %

4. Polymorphism refers to

- a) Multiple functions with same name but different purpose b) Sharing data with child classes
- c) Combining multiple units within one d) Data Hiding

5. Math.round() and Math rint() have return types-

- a) int and double b) double and int
- c) int and int d) double and double

6. Which java package is invoked by default

- a) java.util b) java.lang
- c) java.io d) java.applet

7. To take character as input, we can write

- a) sc.next() b) sc.next().charAt(0)
- b) sc.nextLine() d) sc.nextChar()

8. Math class belongs to

- a) Java.lang
- b) java.util
- b) Java.io
- d) java.Mathematics

9. Extension of byte code is

- a) .byte
- b) .class
- c) .java
- d) .jv

10. Which of the following is used to terminate a construct

- a) break
- b) continue
- c) return
- d) System.exit(0)

11. Which of the following is used to terminate a program

- a) break
- b) continue
- c) System.exit(0)
- d) exit

12. Which type of variable is allowed in switch-case (condition and case value)

- a) int and char
- b) float and double
- c) int, char and String
- d) int and String

13. If float, int and String are involved in an expression, the resultant type would be –

- a) String
- b) float
- c) int
- d) double

14. Which loop executes at least once even if the condition false

- a) for()
- b) while()
- c) do-while()
- d) for-each()

15) if inside if is called

- a) child if
- b) nested if
- c) . if
- c) error

16. Process of creating an object is known as

- a) Initialization
- b) Instantiation

17. Precedence for logical operators is

- | | |
|------------|------------|
| a) ! && | b) && ! |
| b) && ! | d) ! && |

18. The purpose of using \n and \t are

- | | |
|--------------------------------|---------------------------------|
| a) New line and horizontal tab | b) New Line and Text Change |
| b) Horizontal tab and new line | d) new line and carriage return |

19. Dividing a number by 0 returns or finding square root of negative number is

- | | |
|------------------|------------------|
| a) Math error | b) Runtime error |
| c) Logical error | d) Syntax error |

20. Multiple line comment is written within

- | | |
|-----------------|--------------|
| a) // | b) /* ... */ |
| b) <!-- ... --> | d) ## |

SAQ. Answer the following

10x2=20

1. Distinguish between class and object

2. Distinguish between compiler and interpreter

3. int k=5, j=9

```
k+= k++ - ++j +k;  
System.out.println("j=" +j);  
System.out.println("k=" +k);
```

4. double d= 5 +2/3 *6

```
System.out.println(d);
```

5. char ch1= 'a';

```
int n= ch1-32;  
Sytem.out.println(n);  
System.out.println((char)n);
```

6. Write $(a+b)^n / \sqrt{2as + u^2}$ using JAVA

7. find the return type:

```
int a; float f; double k;  
a+b/k (find return type of this expression)
```

8. $(2 > 3 \mid \mid 3 < 4) \&\& 5 != 3$ find the returning boolean value

```
9. for (int i=2; i<=10; i+=2){  
    if( i==2) continue;  
    if(i>7) break;  
    System.out.println(i);  
}
```

```
10. int k=3, i=2;  
    while(i<=5){  
        if(i==2) continue;  
        k*=i;  
    }  
    System.out.println(k);  
Find output and number of times the loop executes.
```

Section B

Write programs in java. Solve any four. Make Value Description Table for each program

15x4=60

1. A shop announces discounts based on purchase amount. Write a program to take purchase amount from user and print discount and amount to be paid.

Purchase Amount	Discount Percentage
Upto 5000	10%
5001 to 10000	15%
10001 to 15000	20%
Above 15000	25%

2. Write a menu driven program to convert user given temperature in Celsius to Fahrenheit and user given temperature in Fahrenheit to Celsius based on user's choice

$$F = (9/5)C + 32$$
$$C = (F - 32) \times 5/9$$

3. Write a menu driven program to print the following based on user's choice.

a) 0 1 1 2 3 5 8 upto n terms (n is given by user) (Fibonacci Series)

b) $S = \frac{a^2}{3} + \frac{a^3}{4} + \frac{a^4}{5} + \frac{a^5}{6} + \frac{a^6}{7} + \frac{a^7}{8} + \frac{a^8}{9} + \dots$ + upto nth term

4. Write a program using switch case to take a number and check

1. if it's a prime number or not
2. find factorial of n

5. Write a program in Java to compute the perimeter and area of a triangle, with its three sides given as a, b, and c using the following formulas:

$$\text{Perimeter} = a + b + c$$

$$\text{Area} = s(s-a)(s-b)(s-c)$$

$$\text{Where } s = (a+b+c) / 2$$

6. Take two numbers a and b from user. Find sum of each even number and sum each odd number between a and b